



JOHN LENN LOPEZ

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PROFESSIONAL SUMMARY

Full-stack developer with hands-on experience building responsive web applications, backend APIs, and embedded system prototypes. Skilled in JavaScript (Vue.js, Electron), PHP (Laravel, CodeIgniter), MySQL, and Docker. Experienced in systems development for government operations, process automation, and queueing systems. Capable of developing Arduino-based IoT and sensor systems through freelance projects. Motivated to contribute to high-quality, scalable solutions in a collaborative engineering environment.

CORE SKILLS

- **AI & Intelligent Systems:** Generative AI Tools (Gemini, ChatGPT), Local LLM Deployment (Ollama open-source model orchestration), Prompt Engineering (context optimization, iterative refinement).
- **Web & Software Development:** Frontend (JavaScript, Vue.js, Electron.js, HTML5, CSS3, Bootstrap, Tailwind CSS); Backend (PHP Laravel, CodeIgniter, Node.js, Python, Microservices); API Architecture (REST APIs, API Development, Authentication & Authorization like OAuth, JWT).
- **Databases & Caching:** MySQL, Redis, Data Modeling, Database Optimization.
- **Tools & DevOps:** Version Control (Git, git-flow, branching, merge conflict resolution), Containerization & OS (Docker, Docker Compose, Ubuntu, FreeBSD), Workflow (UAT Deployment, Google Workspace).
- **Embedded & Hardware (IoT):** Microcontrollers (Arduino ESP32, Uno, Nano, Mega, C/C++ for microcontrollers, Raspberry Pi), Hardware Integration (sensor integration for ultrasonic, turbidity, RFID, temperature, serial communication, IoT prototyping).
- **Systems & Architecture:** System flowcharts, Entity-Relationship Diagrams (ERD) creation, technical documentation, user training, system demonstrations.

WORK EXPERIENCE

Department of Social Welfare and Development MIMAROPA – Malate, Manila

Programmer III (Full-Stack Developer) | August 16, 2024 – Present

- Improved internal applications from UI to database layer, implementing role-based access, optimized queries, and new modules.
- Designed system flows, ERDs, and UI mock-ups for newly requested features.
- Performed deployment to staging and UAT environments, resolved integration issues, and supported multi-team testing.
- Developed and enhanced responsive systems for employee attendance, accomplishment tracking, and paperless workflows.
- Collaborated with business owners to translate requirements into functional system enhancements.

Department of Social Welfare and Development MIMAROPA – Malate, Manila

Computer Programmer II (Full-Stack Developer) | June 2023 – August 2024

- Developed inventory and asset management systems for laptops and accountable equipment.
- Configured operating systems for use and established secure network connections.

JRA Technology Inc. – Makati City

Junior Software Engineer / System Administrator | August 2022 – June 2023

- Developed inventory and asset management systems for laptops and accountable equipment.
- Configured operating systems for use and established secure network connections.

FREELANCE EXPERIENCE

Freelance Software & Embedded Systems Developer

Independent Contractor | 2020 – Present

- Designed and built custom Arduino projects, including automation modules, IoT sensors, monitoring systems, and RFID-based devices.
- Developed embedded logic using C/C++ for Arduino boards and ESP32 modules.
- Built API endpoints for IoT projects and performed serial, WiFi, and Bluetooth communication testing.
- Created wiring diagrams, documentation, and technical project demonstrations for clients.
- Delivered end-to-end solutions spanning requirements gathering, coding, assembly, testing, and field deployment.

MAJOR PROJECTS

- **SSO MIMAROPA PORTAL:** Full-stack development using Laravel Inertia React.js; implemented SSO Keycloak and LDAP/AD connection for single sign-on to all microservices accessed by employees.
- **FACED IS:** Family Access Card in Emergencies and Disasters Information System. Digital tool for profiling disaster-affected families and identifying assistance services. Built with RESTful API Laravel and Vue.js; implemented dashboards, automated reporting, and role management.
- **Performance Management Information System (DSWD):** Full-stack development using RESTful API Laravel and Vue.js; engineered dashboards, automated reporting, and supervisor monitoring features to secure transparency of annual performance.
- **Daily Time Record Management Information System (DSWD):** Full-stack development using RESTful API Laravel and Vue.js; automated logs management, dashboard metrics, and reporting pipelines.
- **Queueing and Beneficiary Management System (DSWD):** Designed and implemented a real-time queueing platform for social workers, significantly increasing client service operational velocity.
- **Travel Expense Monitoring Information System (DSWD):** Full-stack development using Laravel and Vue.js; built tracking pipelines for administrative process signatory loops among section chiefs and heads.
- **IoT Water Quality Monitoring System (Freelance):** Developed a hardware prototype using Arduino & ESP32 for turbidity and temperature measurement that synchronizes real-time data to a RESTful API Node.js backend dashboard.
- **PC Inventory System (DSWD):** Full-stack development using Laravel and Vue.js; automated lifecycle logs and accountability tracking for employee laptops and desktops.

CERTIFICATIONS

- Online Refresher Course on Google Workspace (2025) – DSWD
- Information Security and Data Protection (2025) – DSWD
- Data Visualization and Dashboard Creation: Tableau Desktop & Prep (2024) – DSWD
- Cybersecurity and Data Protection (2023) – DSWD
- Cybersecurity (2023) – DSWD
- Software Engineering Framework (2023) – DSWD

EDUCATION

STI College Sta. Maria

Bachelor of Science in Computer Engineering | 2018 – 2022

Cumulative GWA: 1.81

STI College Sta. Maria

Senior High School – STEM Strand | 2016 – 2018

PROFESSIONAL REFERENCES

Roselyn G. Magdaraog

Business Analyst, DSWD

MIMAROPA

Phone: 09954645603

Glenda Morales

Assistant Professor, STI College

Email:

glenda.morales@stamaria.sti.ph

Jann Bezzli M. Andal

IT Engineer / Business Analyst

Phone: 09394309904

I hereby certify that the above information given are true and correct as to the best of my knowledge and belief


John Lenn. Lopez
Applicant